

Dipal Katuwal

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PROFESSIONAL SUMMARY

Full Stack Engineer delivering end-to-end web applications, ML-powered tools, and GenAI-integrated workflows. Proficient in JavaScript and Python across the full stack, with a strong focus on clean architecture and rapid iteration using modern AI tooling. Known for taking full ownership of features, collaborating effectively across teams, and consistently delivering under tight timelines.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL, HTML/CSS
Frameworks: Next.js, React, Node.js, Express, Django REST Framework
Machine Learning: TensorFlow, Keras, LSTM Neural Networks, Scikit-learn, Pandas, NumPy
Backend & DevOps: MongoDB, PostgreSQL, REST APIs, JWT, Zod, Git, Vercel, Docker, CI/CD
UI/UX: Tailwind CSS, Shadcn UI, Framer Motion, Material UI

PROFESSIONAL EXPERIENCE

Full Stack Developer Intern

July 2025 – Oct 2025

Need Technosoft Pvt. Ltd.

Biratnagar, Nepal

- Developed full-stack features including a user-facing dashboard and backend data processing endpoints using Node.js and Express; optimized database queries and introduced request caching to improve API performance on high-traffic routes.
- Migrated 10+ legacy UI components to responsive Tailwind CSS layouts and collaborated in an agile team through code reviews and weekly sprints.

FEATURED PROJECTS

NepseSage | Next.js 15, Express, MongoDB, TypeScript, Zod | [Live Demo](#)

- Built a trading psychology tracking system to analyze user discipline patterns and behavioral biases.
- Architected an Express backend with JWT security, Zod validation, and Node-cron for automated data synchronization.

NEPSE Prediction Portal | Python, Django REST Framework, TensorFlow, React

- Developed an end-to-end predictive pipeline utilizing LSTM Neural Networks to forecast price trends for 100+ NEPSE tickers.
- Integrated a Django REST API to serve model predictions and technical analysis indicators to a React dashboard.

PasalBot | React, Vite, Context API, Tailwind CSS

- Designed a modular commerce platform for social media sellers, featuring automated order tracking and inventory management.
- Managed global state using Context API and useReducer to improve frontend responsiveness and usability.

EDUCATION

Mahendra Morang Adarsha Multiple Campus

Biratnagar, Nepal

BSc in Computer Science and Information Technology (CSIT)-TU

2021 – 2026

- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Artificial Intelligence, Software Engineering.